



Causal sufficient dimension reduction for multiple continuous exposures

an application to environmental mixtures analysis

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Introduction

Goals of Mixtures Analysis

Evaluating the health effects of complex environmental mixtures has become a central priority for the NIEHS.

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- Assess the joint health impact
- Identify important contributors
- Inform policy regulations

Environmental mixtures analysis methods

Single-index methods ($d = 1$):

- Weighted quantile sum regression (WQS)
- Quantile g -computation (qgcomp)

Dimension reduction methods ($1 < d < p$):

- Principal components pursuit regression (PCP)
- Bayesian multiple index models (BMIM)

Full exposure-response surface (ERS) methods($d = p$):

- Bayesian kernel machine regression (BKMR)
- Bayesian additive regression trees (BART)
- Distributed lag models with regression trees (DLM tree)

Limitations of existing methods

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$$Y = h(X) + C^{\top} \gamma + \epsilon$$

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1. The truth might lie between the single index and full ERS paradigms.
2. Many dimension reduction methods don't take into account the outcome Y
3. None of these (except for qgcomp sometimes) are *causal inference* methods.

Proposed solution

We propose a method, **causal sufficient dimension reduction**, that is simultaneously *causal*, *flexible*, and *interpretable*.

Causal inference

- Observed data: $O = (Y, X, C) \in \mathcal{Y} \times \mathcal{X} \times \mathcal{C}$
 - $Y \in \mathbb{R}$: continuous outcome
 - $X \in \mathbb{R}^p$: continuous exposures
 - $Z \in \mathbb{R}^d$: latent exposures
 - $C \in \mathbb{R}^q$: covariates
- Data: i.i.d. sample $(O_1, \dots, O_n)$
- Potential outcomes: $Y^x = \text{outcome under exposure } x$
- Causal mean with intervention x : $\mu(x) := \mathbb{E}[Y^x]$
- Outcome regression: $m(x, c) := \mathbb{E}(Y \mid X = x, C = c)$
- Propensity score: $\pi(x \mid c) = \partial P(X \leq x \mid C = c) / \partial x$

Doubly robust estimation of causal ERS

Colangelo & Lee (2025) and Zhang & Chen (2025) provide a $\sqrt{nh}$ -convergent doubly-robust estimator for the dose-response curve for univariate continuous treatments.

We extend to the multivariate case as follows:

Doubly-robust estimator of $\mu(x)$

$$\hat{\mu}_{DR}(x) = \frac{1}{nh^p} \sum_{i=1}^n \left\{ \frac{K\left(\frac{X_i - x}{h}\right)}{\hat{\pi}(X_i | C_i)} \cdot [Y_i - \hat{m}(x, C_i)] + h^p \cdot \hat{m}(x, C_i) \right\}.$$

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Fitting a $(p + q)$ -regression and a p -variate conditional density with q regressors can be hard!

Can we find a low rank Z that preserves all the information about $\mu(x)$ contained in X ?

Sufficient dimension reduction

Sufficient Dimension Reduction (SDR)

Goal: find a linear transformation $\beta \in \mathbb{R}^{p \times d}$ that retains all information about $\mathbb{E}(Y | X)$ in X .

$$\mathbb{E}(Y | X = x) = \mathbb{E}(Y | \beta^\top X = \beta^\top x)$$

We refer to $\text{span}(\beta)$ as the **central mean subspace (CMS)** denoted by $\mathcal{S}_{\mathbb{E}(Y|X)}$.

Semiparametric Estimation of the CMS

The efficient score for β (Luo and Cai, 2016) is:

$$U(\beta) = \frac{1}{\sigma^2(X)} \left[X - \frac{\mathbb{E}\{X/\sigma^2(X) \mid \beta^\top X\}}{\mathbb{E}\{1/\sigma^2(X) \mid \beta^\top X\}} \right] \\ \times \frac{\partial \mathbb{E}(Y \mid \beta^\top X)}{\partial(\beta^\top X)} [Y - \mathbb{E}(Y \mid \beta^\top X)]$$

Our estimate $\hat{\beta}$ solves $U(\hat{\beta}) = 0$ by estimating nuisances:

1. $\sigma^2(X) := \text{Var}(Y \mid X)$
2. $\mathbb{E}\{X/\sigma^2(X) \mid \beta^\top X\}$
3. $\mathbb{E}\{1/\sigma^2(X) \mid \beta^\top X\}$
4. $\partial \mathbb{E}(Y \mid \beta^\top X) / \partial(\beta^\top X)$
5. $\mathbb{E}(Y \mid \beta^\top X)$

The minimum average variance estimator (MAVE) from Xia et al. (2002) achieves estimation of CMS through a variation of local linear regression.

Model:

$$Y = g(\beta^\top X) + \varepsilon, \quad \mathbb{E}(\varepsilon \mid X) = 0$$

MAVE objective:

$$\min_{\beta^\top \beta = I_d, \{a_j, b_j\}} \sum_{j=1}^n \sum_{i=1}^n \left(Y_i - \{a_j + b_j^\top \beta^\top (X_i - X_j)\} \right)^2 w_{ij}$$

where $w_{ij} = K_h(X_i - X_j) / \sum_{\ell=1}^n K_h(X_i - X_\ell)$.

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How can we bridge SDR for $\mathbb{E}(Y \mid X)$, with the doubly robust estimator $\hat{\mu}_{DR}(x)$ for the ERS from earlier?

Causal sufficient dimension reduction

Suppose that there exists $\beta \in \mathbb{R}^{p \times d}$

$$\mu(x) = \mu_{\beta}(\beta^{\top} x), \quad x \in \mathcal{X}.$$

For causal SDR, our new estimand of interest is the **causal central mean subspace** (CCMS) denoted as $\mathcal{S}_{\mathbb{E}(Y|X)}$.

Semiparametric Estimation in the Causal Setting

Using the theory of marginal structural models (Robins, 2000), Nabi et al. (2022) proposed the following locally efficient score for estimating $\mathcal{S}_{\mathbb{E}(Y^x)}$.

$$\begin{aligned}\tilde{U}(\beta) &= \mathbb{E}_q[\mathbb{E}[U^*(\beta) \mid X, C] \mid C] \\ &\quad + \frac{\pi^*(X)}{\pi(X \mid C)} \left(U^*(\beta) - \mathbb{E}[U^*(\beta) \mid X, C] \right)\end{aligned}$$

where U^* is the MSM version of the association-based score $U(\beta)$ and $\mathbb{E}_q(\cdot)$ denotes the *causal* mean.

Total number of nuisance functions

The number of nuisances required to evaluate $\tilde{U}(\beta)$ is much larger.

1. $\text{Var}_q(Y \mid X)$
2. $m(x, c)$
3. $\pi(x \mid c)$
4. $\mathbb{E}_q[\cdot \mid C]$
5. $\mathbb{E}_q\{X/\sigma^2(X) \mid \beta^\top X\}$
6. $\mathbb{E}_q\{1/\sigma^2(X) \mid \beta^\top X\}$
7. $\partial \mathbb{E}_q(Y \mid \beta^\top X) / \partial (\beta^\top X)$
8. $\mathbb{E}_q(Y \mid \beta^\top X)$

Is there a way to conduct causal SDR that avoids such heavy β -dependent nuisance estimation while attaining acceptable efficiency?

Multiple options for Causal SDR

New strategy

Find new variables $\tilde{Y}$ and $\tilde{X}$ such that $\mathbb{E}(\tilde{Y} \mid \tilde{X} = x) = \mu_{\beta}(\beta^{\top} x)$. This allows us to take advantage of standard SDR methods.

Multiple options for Causal SDR

New strategy

Find new variables $\tilde{Y}$ and $\tilde{X}$ such that $\mathbb{E}(\tilde{Y} \mid \tilde{X} = x) = \mu_{\beta}(\beta^{\top} x)$. This allows us to take advantage of standard SDR methods.

Method	Y variable	X variable
Oracle	$\mu(X_i)$	X_i
Regression adjustment (RA)	$\hat{\mu}_{RA}(X_i)$	X_i
Doubly robust (DR)	$\hat{\mu}_{DR}(X_i)$	X_i
Pseudo-outcome (PO)	$\hat{\xi}_i$	X_i
Residualized pair (RP)	$Y_i - \hat{\mathbb{E}}(Y \mid C_i)$	$X - \hat{\mathbb{E}}(X \mid C_i)$

NW leave-one-out estimator

$$\hat{a}_{d0,j} = \frac{\sum_{i \neq j} K_{h_d}^{(i,j)} \tilde{Y}_i}{\sum_{i \neq j} K_{h_d}^{(i,j)}}, K_{h_d}^{(i,j)} = K_{h_d} \{ \hat{\beta}_1^\top (\tilde{X}_i - \tilde{X}_j), \dots, \hat{\beta}_d^\top (\tilde{X}_i - \tilde{X}_j) \}$$

Cross-validated error

$$\text{CV}_n(d) = \frac{1}{n} \sum_{j=1}^n (\tilde{Y}_j - \hat{a}_{d0,j})^2, \quad \text{CV}_n(0) = \frac{1}{n} \sum_{i=1}^n (\tilde{Y}_i - \bar{Y})^2$$

Estimated structural dimension

$$\hat{d} = \arg \min_{0 \leq d \leq p} \text{CV}_n(d)$$

Summary of causal SDR procedure

1. Estimate new response and exposure variables $(\tilde{Y}, \tilde{X})$
2. Input the new $(\tilde{Y}, \tilde{X})$ into standard MAVE.
3. Estimate structural dimension with $\hat{d}$.
4. Apply Newton-Raphson update to estimate $\mathcal{S}_{\mathbb{E}(Y^x)}$.

What are the statistical properties of MAVE when

1. Using noisy estimated versions of the true Y and X ?
2. When $\epsilon = 0$ (since $\hat{\mu}(x)$ converges to a deterministic quantity)?

Causal MAVE convergence

Theorem (Convergence of csMAVE)

Under regularity assumptions, if $h \rightarrow 0$, $d \geq d_0$, and $nh^p / \log(n) \rightarrow \infty$, then

$$\|(I_p - \hat{\beta}\hat{\beta}^\top)\beta\| = O_p(h^3 + h\delta_n + h^{-1}\delta_n^2 + \textcolor{red}{h}^{-1}R_n)$$

where $\delta_n = \{\log n / (nh^p)\}^{1/2}$ and $\max_{1 \leq i \leq n} |\hat{\mu}(X_i) - \mu(X_i)| = O_p(R_n)$.

The MAVE convergence rate is maintained so long as R_n is fast enough (if d_0 is small and $m(x, c)$ and $\pi(x | c)$ are estimated well).

Theorem (Consistency of $\hat{d}$)

The estimator $\hat{d}$ converges to the structural dimension d_0 in probability as $n \rightarrow \infty$.

Theorem (Post dimension reduction inference)

Let $\sqrt{n}(\hat{\beta}_n - \beta_0) = O_P(1)$ and $\hat{\mu}_{DR}(x; \beta)$ be the DR estimator using the intervention $\beta^\top X$. Under assumptions, if $h \rightarrow 0$ and $nh^d \rightarrow \infty$ as $n \rightarrow \infty$, then $\hat{\mu}_{DR}(x_0; \hat{\beta}_n)$ consistently estimates $\mu(x_0)$. The estimator based on the estimated reduction is asymptotically equivalent to the oracle estimator:

$$\sqrt{nh_n^d} \{ \hat{\mu}_{DR}(x_0; \hat{\beta}_n) - \hat{\mu}_{DR}(x_0; \beta_0) \} = o_P(1)$$

Plugging in $\hat{\beta}$ into the DR estimator gives no penalty (asymptotically) compared to the true β_0 .

Why not interpret β directly?

Suppose $d_0 = 2$.

Basis representation 1

$$\beta = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Naive interpretation:

- X_1 important
- X_2 important
- X_3 unimportant

Basis representation 2 (same subspace)

$$\tilde{\beta} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 \end{pmatrix}$$

Columns remain **orthonormal**, and

$$\text{span}(\beta) = \text{span}(\tilde{\beta})$$

But the entries suggest **different importance** for X_1 and X_2 .

Key point

The individual coefficients of β **depend on the chosen orthonormal basis.**

Instead we define the **Subspace Importance Score**

Subspace Importance Score

We define a measure of variable importance called the **Subspace Importance Score (SIS)**.

Let

$$P = \beta\beta^\top$$

be the **projection matrix onto the central subspace** $\text{span}(\beta)$.

For exposure X_j , define

$$\lambda_j = P_{jj}.$$

Properties

- For each $j = 1, \dots, p$,

$$0 \leq \lambda_j \leq 1.$$

- The scores sum to the **structural dimension**:

$$\sum_{j=1}^p \lambda_j = d.$$

- λ_j depends only on the **subspace $\text{span}(\beta)$** , and is invariant to the particular basis used for β .

Interpretation

λ_j measures **how strongly exposure X_j contributes to the CCMS**.

- $\lambda_j = 1 \rightarrow X_j$ is completely present in the true ERS
- $\lambda_j = 0 \rightarrow X_j$ has no role in the true ERS

Simulation Analysis

Simulation Setup (Part 1)

Data Generating Process

- **Exposures & Confounders:** $p = 10$ exposures (X) with AR(1) correlation ($\rho = 0.8$) and $q = 5$ independent normal confounders (C).
- **True Subspace:** The structural dimension is $d_0 = 2$, with β solely spanning X_1 and X_2 .
- **Outcome Model:** $Y = h(Z) + g(C) + f(Z, C) + \epsilon$
 - $h(Z) = 4 \sin(Z_1) + 2Z_2^2 + Z_1Z_2$.
 - $g(C) = 5.0 \tanh(C_1) + 5.0(C_2^2 - 1) + 5.0(C_3^2 - 1) + 5.0 \sin(1.5C_4)$
 - $f(Z, C) = 5C_5(Z_1 + Z_2)$

Nuisance Function Estimation

- $\mathbb{E}[Y \mid X, C]$: 2-layer neural network via `torch` using 150 epochs, a 0.05 learning rate, and SiLU activation functions.
- $\pi(x \mid c)$: Linear multivariate normal (MVN) density estimation.
- $\mathbb{E}[Y \mid C], \mathbb{E}[X \mid C]$: SuperLearner ensemble combining GLM, Lasso (`glmnet`), and Multivariate Adaptive Regression Splines (`earth`).

Comparison methods

Method	Estimation of the Central Mean Subspace (β)
PCA	Top eigenvectors of $\text{Cov}(\mathbf{X})$ (unsupervised).
pCCA	Maximizes linear correlation between $\mathbf{X}$ and Y after linearly partialling out $\mathbf{C}$.
MAVE	MAVE applied to (Y, X) .
Oracle-MAVE	MAVE applied to $(\mu(X), X)$.
RA-MAVE	MAVE applied to $(\hat{\mu}_{RA}(X), X)$.
DR-MAVE	MAVE applied to $(\hat{\mu}_{DR}(X), X)$.
PO-MAVE	MAVE applied to $(\hat{\xi}, X)$.
RP-MAVE	MAVE applied to $(Y - \hat{\mathbb{E}}[Y C], X - \hat{\mathbb{E}}[X C])$.
Full $\mathbf{X}$	No dimension reduction. $\hat{\beta} = I_p$.

We want to know how well we estimate

1. $\mathcal{S}_{\mathbb{E}(Y^x)}$: the causal central mean subspace
2. d_0 : the structural dimension
3. $\mu_\beta(z)$: the causal ERS, using $\hat{Z} = \hat{\beta}^\top X$ in place of X .

Results

Table 1: Mean (SD) Frobenius norm error between estimated and true $\mathcal{S}_{Y^x}$ when d_0 is known.

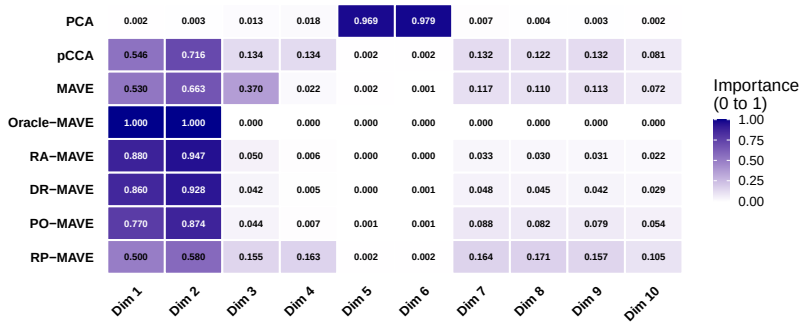
Method	$n = 100$	$n = 500$	$n = 1000$	$n = 2500$	$n = 5000$
PCA	1.997 (0.002)	1.997 (0.001)	1.997 (0.001)	1.997 (0.000)	1.997 (0.000)
pCCA	1.316 (0.119)	1.202 (0.177)	1.100 (0.202)	0.925 (0.219)	0.743 (0.203)
MAVE	1.468 (0.220)	1.247 (0.247)	1.239 (0.229)	1.281 (0.179)	1.299 (0.148)
Oracle-MAVE	0.203 (0.084)	0.030 (0.010)	0.015 (0.005)	0.006 (0.002)	0.003 (0.001)
RA-MAVE	1.276 (0.331)	0.442 (0.388)	0.216 (0.272)	0.072 (0.152)	0.026 (0.027)
DR-MAVE	1.458 (0.253)	0.546 (0.352)	0.297 (0.245)	0.137 (0.138)	0.079 (0.066)
PO-MAVE	1.486 (0.315)	0.691 (0.484)	0.450 (0.433)	0.292 (0.368)	0.230 (0.317)
RP-MAVE	1.658 (0.173)	1.339 (0.220)	1.171 (0.263)	0.907 (0.332)	0.689 (0.340)

Order Determination

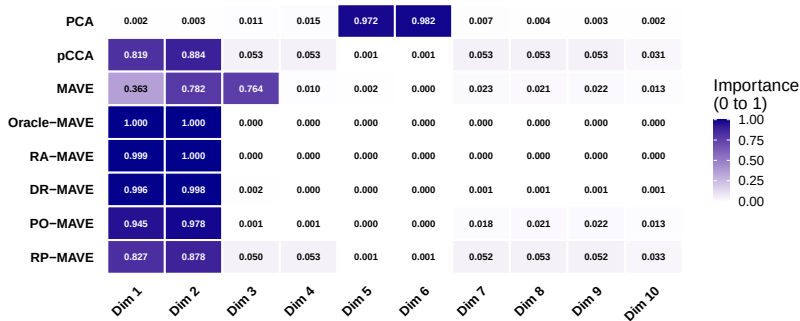
Table 2: Proportion of simulations selecting each structural dimension.

n	Method	$d = 0$	$d = 1$	$d = 2$	$d = 3$	$d = 4$	$d = 5$
500	MAVE	0.00	0.00	0.11	0.75	0.14	0.00
	Oracle-MAVE	0.00	0.00	1.00	0.00	0.00	0.00
	RA-MAVE	0.01	0.03	0.88	0.07	0.01	0.00
	DR-MAVE	0.01	0.05	0.85	0.07	0.01	0.00
	PO-MAVE	0.03	0.06	0.73	0.14	0.03	0.00
	RP-MAVE	0.00	0.28	0.29	0.23	0.15	0.04
5000	MAVE	0.00	0.00	0.00	1.00	0.00	0.00
	Oracle-MAVE	0.00	0.00	1.00	0.00	0.00	0.00
	RA-MAVE	0.00	0.00	1.00	0.00	0.00	0.00
	DR-MAVE	0.00	0.00	1.00	0.00	0.00	0.00
	PO-MAVE	0.01	0.00	0.92	0.07	0.00	0.00
	RP-MAVE	0.00	0.21	0.74	0.05	0.00	0.00

Subspace Importance Scores ($n = 500$)



Subspace Importance Scores ($n = 5000$)



ERS Pointwise Bias

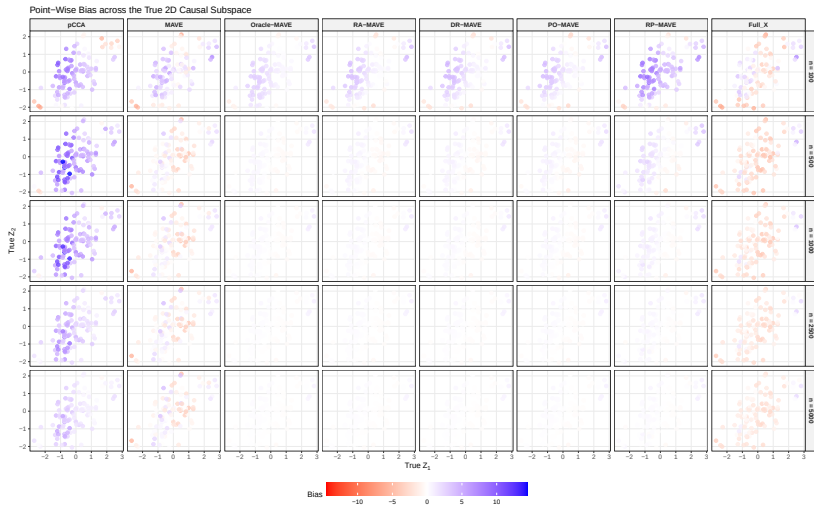


Figure 1: Bias of ERS estimation.

ERS Pointwise RMSE

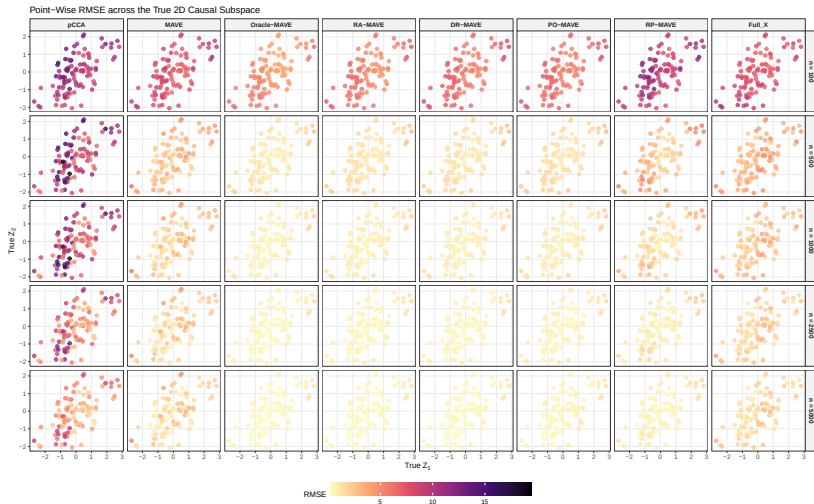


Figure 2: RMSE of ERS estimation.

ERS Pointwise Coverage

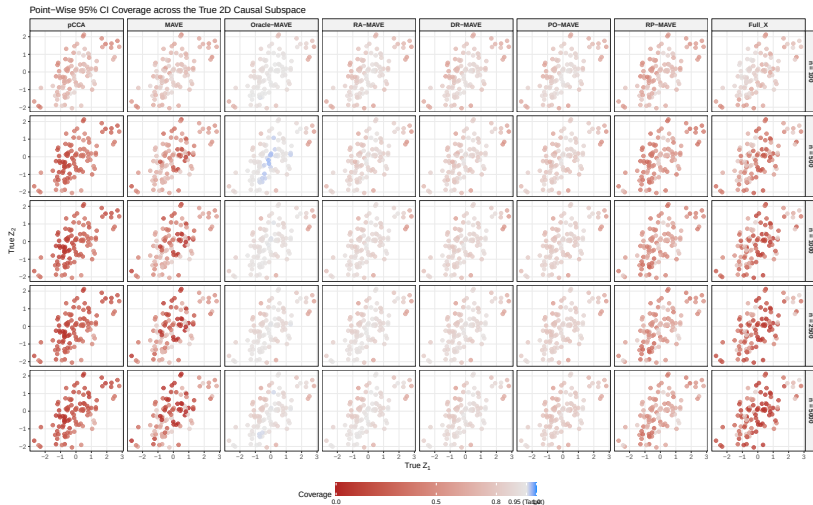


Figure 3: Coverage of 95% pointwise CIs.

Real data analysis

Study population: $N = 303$ African-American women in Atlanta from 2014-2020 with term births.

Exposure: Four log2-transformed PFAS exposures ($p=4$): PFOS, PFOA, PFNA, PFHxS.

Covariates: Maternal age, BMI, education, tobacco use, marijuana use, and infant sex.

Outcome: Birthweight in grams.

Subspace Importance Scores

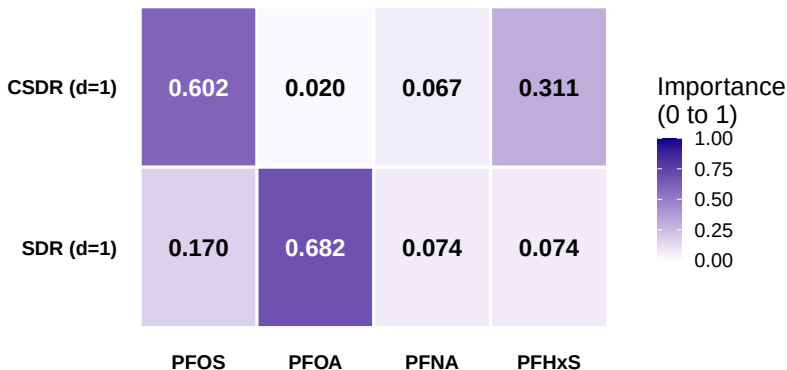


Figure 4: Subspace importance scores between causal SDR and association-based SDR

Estimate of the ERS

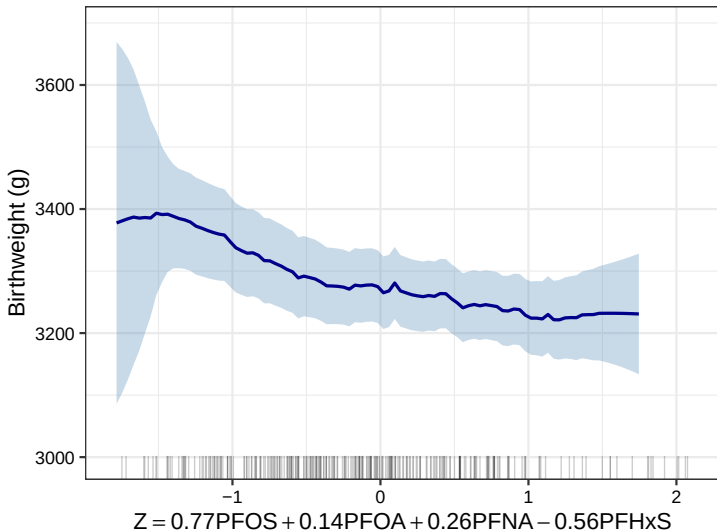


Figure 5: Causal ERS estimate with 95% CIs.

Summary of causal SDR

We have designed a method that:

1. Implements dimension reduction for X that preserves the causal dose-response (Theorem 1)

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3. Improves efficiency and coverage over using the original X and incurs no asymptotic plug-in bias for ERS estimation (Theorem 3)
4. Leverages existing off-the-shelf tools for SDR
5. Gives clear outputs for causal ERS estimation and variable importance

Acknowledgments

Thank you to my postdoc advisers Drs. Razieh Nabi and Howard Chang, and collaborators Drs. Donghai Liang and Anne Dunlop for access to the ATL-AA cohort data!

Questions?

Appendix

Causal Identification Assumptions

1. **Consistency:** If $X = x$, then the observed outcome equals the potential outcome: $Y = Y^x$
2. **Positivity:** Every exposure level has nonzero probability (given covariates): $\pi(\mathbf{x} \mid \mathbf{c}) \geq \varepsilon > 0$
3. **Conditional ignorability:** Given covariates, exposure is independent of potential outcomes: $Y^{\mathbf{x}} \perp\!\!\!\perp \mathbf{X} \mid \mathbf{C}$

Remark (Positivity)

We make the positivity assumption here for identification purposes rather than assuming its validity in environmental applications. However, identifying a low rank Z can make positivity more tenable than for X .

Causal response variables

Recall again the underlying model:

$$\mu(x) = g(\beta^\top x), \quad x \in \mathcal{X}.$$

Notice that if $\mathbb{E}(W \mid X = x) = \mu(x)$, then plugging (W, X) into the MAVE yields a valid estimate of the CCMS.

RA and DR responses

One approach is to estimate $\mu(x)$ at $X_1, X_2, \dots, X_n$ to yield $\hat{\mu}(X_1), \dots, \hat{\mu}(X_n)$.

This can be achieved with the mentioned RA and DR approaches.

Pseudo-outcomes (PO)

By a multivariate extension of Kennedy et al. (2017), a causal pseudo-outcome can be designed:

$$\xi_j = \frac{Y_j - m(X_j, C_j)}{\pi(X_j | C_j)} \left\{ \frac{1}{n} \sum_{i=1}^n \pi(X_j | C_i) \right\} + \frac{1}{n} \sum_{i=1}^n m(X_j, C_i)$$

which satisfies

$$\xi = \mu_{\beta}(\beta^{\top} X) + \epsilon_{\xi}, \quad \mathbb{E}(\epsilon_{\xi} | X) = 0.$$

Residualized Pair (RP)

The RA, DR, and PO approaches all require estimation of $m(x, c)$ and $\pi(x | c)$, which may be unappealing if the goal is dimension reduction.

If we are willing to assume no interaction between X and C in $m(x, c)$:

Proposition

Let $\tilde{Y} = Y - \mathbb{E}(Y | C)$ and $\tilde{X} = X - \mathbb{E}(X | C)$. Under additive confounding, for some smooth function $\tilde{g}$,

$$\mathbb{E}(\tilde{Y} | \tilde{X}) = \tilde{g}(\beta^\top \tilde{X}).$$

Comparison between MAVE and EE estimators

Table 3: Mean (SD) of Frobenius norm error between the estimated and true subspace between MAVE and one-step EE.

Method	$n = 100$		$n = 500$		$n = 1000$		$n = 2500$		$n = 5000$	
	MAVE	EE	MAVE	EE	MAVE	EE	MAVE	EE	MAVE	EE
Oracle	0.216 (0.136)	0.635 (0.261)	0.030 (0.010)	0.379 (0.131)	0.015 (0.005)	0.316 (0.107)	0.006 (0.002)	0.254 (0.086)	0.003 (0.001)	0.219 (0.070)
RA	1.343 (0.341)	1.352 (0.332)	0.447 (0.392)	0.463 (0.384)	0.223 (0.288)	0.232 (0.282)	0.073 (0.157)	0.080 (0.153)	0.026 (0.035)	0.031 (0.035)
DR	1.537 (0.267)	1.543 (0.262)	0.568 (0.369)	0.589 (0.361)	0.303 (0.257)	0.323 (0.253)	0.140 (0.151)	0.162 (0.148)	0.078 (0.058)	0.100 (0.061)
PO	1.487 (0.318)	1.491 (0.315)	0.663 (0.450)	0.657 (0.445)	0.432 (0.401)	0.412 (0.393)	0.287 (0.345)	0.246 (0.333)	0.221 (0.293)	0.165 (0.280)
RP	1.768 (0.218)	1.776 (0.217)	1.413 (0.259)	1.443 (0.254)	1.192 (0.240)	1.226 (0.237)	0.913 (0.255)	0.952 (0.249)	0.676 (0.285)	0.697 (0.271)

Assumption for csMAVE

Assumption (csMAVE)

1. (Y_i, X_i, C_i) are i.i.d.
2. X and $\mu(X)$ have finite moments of all orders.
3. $f_X(x)$ has bounded fourth derivatives and is bounded away from zero on compact support $\mathcal{X}$.
4. $\mu(X) = g(\beta^\top X)$ with $\beta^\top \beta = I_d$ and g having bounded continuous third derivatives.
5. $K(\cdot)$ is bounded, spherically symmetric, differentiable, and satisfies $\int UU^\top K(U) dU = I$.
6. $\max_{1 \leq i \leq n} |\hat{\mu}(X_i) - \mu(X_i)| = O_p(R_n)$.